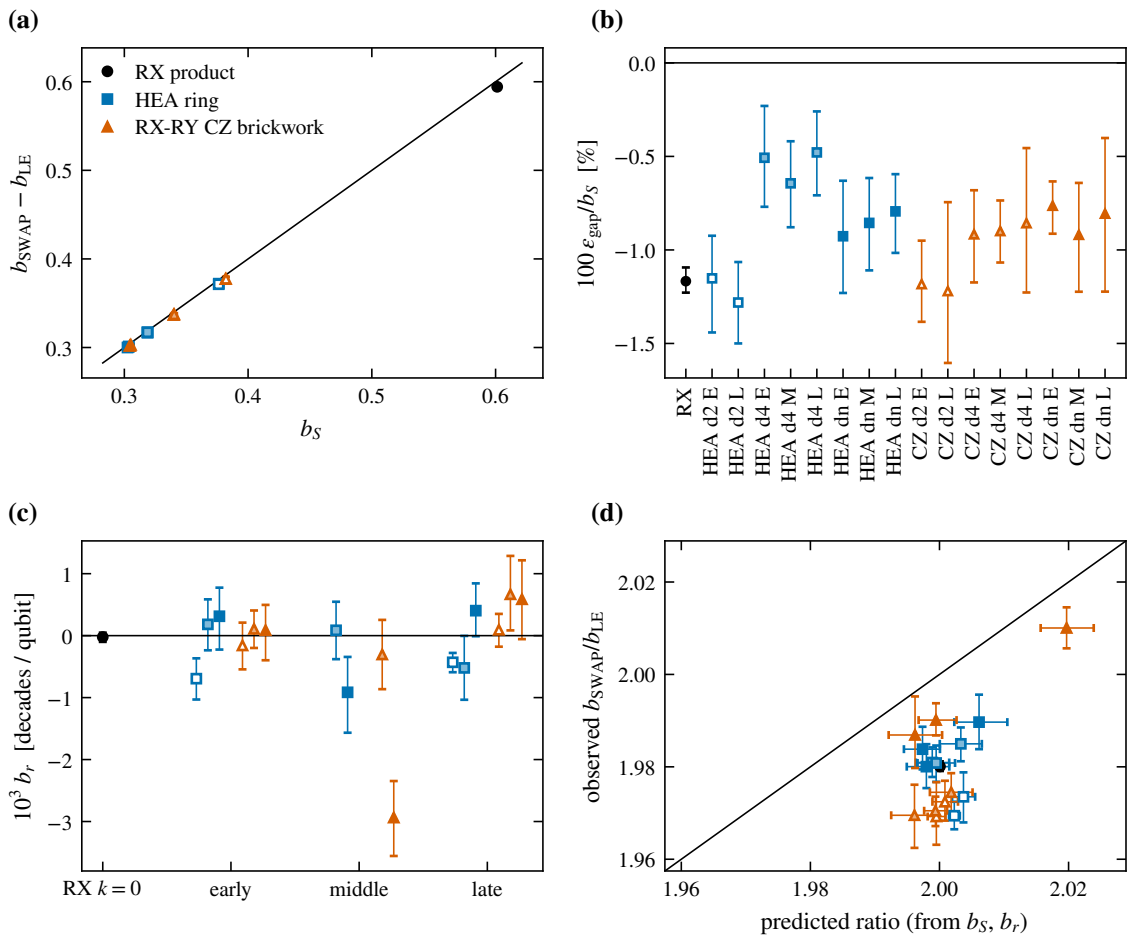
(a)
RX product
HEA ring
RX-RY CZ brickwork
$b_{SWAP} - b_{LE}$
b_S
0.3
0.4
0.5
0.6
(b)
$100\,\varepsilon_{gap}/b_S$ [%]
0.0
−0.5
−1.0
−1.5
RX
HEA d2 E
HEA d2 L
HEA d4 E
HEA d4 M
HEA d4 L
HEA dn E
HEA dn M
HEA dn L
CZ d2 E
CZ d2 L
CZ d4 E
CZ d4 M
CZ d4 L
CZ dn E
CZ dn M
CZ dn L
(c)
$10^3\,b_r$ [decades / qubit]
1
0
−1
−2
−3
RX $k = 0$
early
middle
late
(d)
observed b_{SWAP}/b_{LE}
2.02
2.00
1.98
1.96
predicted ratio (from b_S, b_r)